MODULE 1 · LECTURE 5

MS Office Applications: Word, Excel and PowerPoint

Creating, editing and formatting documents; calculating, analysing and graphing data; and presenting results — with farm-record, rainfall and sensor-calibration examples.

Learning objectives

After this lecture you should be able to:

1. Create, edit and format a professional document in MS Word, using styles, tables, references and mail merge.
2. Organise data in MS Excel and write formulas with relative and absolute cell references.
3. Use common Excel functions — SUM, AVERAGE, MAX, COUNTIF, IF and XLOOKUP — on farm data.
4. Choose and build the right chart, and add a trendline for calibration data.
5. Prepare a clear presentation in MS PowerPoint for a seminar or farmers' meeting.
6. Recognise Office file formats and free alternatives.

Scope: Lecture 4 covered the operating system on which these programs run. This lecture introduces the three most-used office applications. Databases (MS Access and DBMS) follow in Lecture 6. Practise each example on a computer — these are skills, learnt by doing.

1. Office suites

An **office suite** is a package of application programs for everyday office work. Microsoft Office (now sold mainly as Microsoft 365) includes **Word** (word processing), **Excel** (spreadsheets), **PowerPoint** (presentations), **Access** (databases) and **Outlook** (e-mail). Free alternatives include **LibreOffice** (open-source) and **Google Docs, Sheets and Slides** (in a web browser). The skills in this lecture carry over to all of them.

Extension	Program	Standard
.docx, .xlsx, .pptx	Word, Excel, PowerPoint	Office Open XML — ECMA-376 (2006), ISO/IEC 29500 (2008) (Microsoft, n.d.-d)
.odt, .ods, .odp	LibreOffice Writer, Calc, Impress	OpenDocument Format — ISO/IEC 26300
.pdf	Any (export or print)	Portable Document Format — keeps the layout fixed for sharing and printing
.csv	Any spreadsheet	Plain-text, comma-separated values — the usual export format of data loggers

2. MS Word — word processing

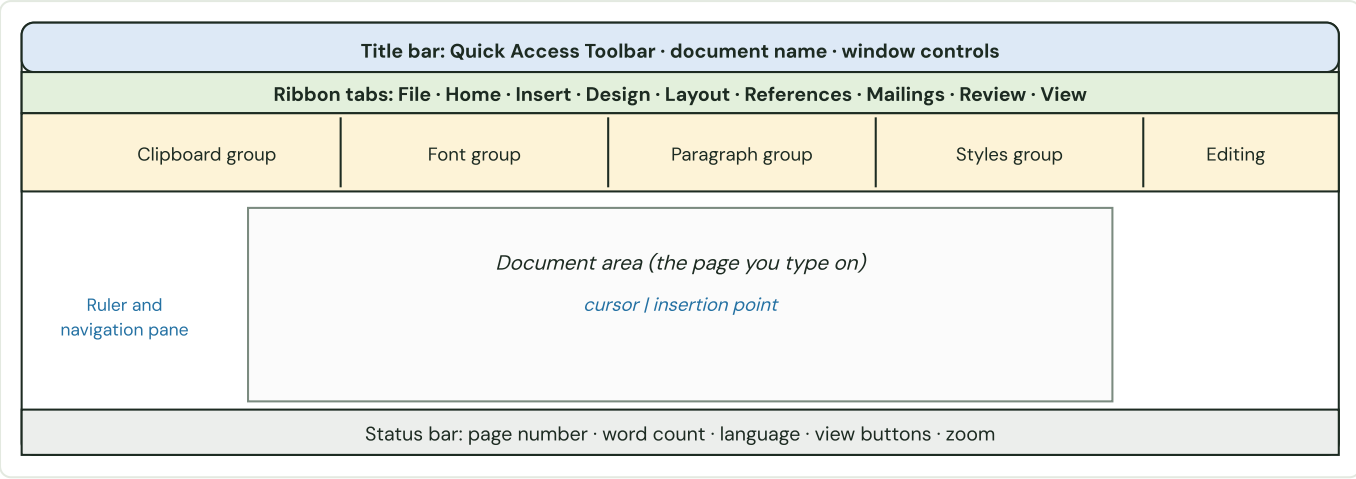


Figure 1. Main parts of the MS Word window (schematic). Original diagram.

The features below follow Microsoft's own Word help pages (Microsoft, n.d.-a); practise each one as you read.

2.1 Creating and saving

- **New** — a blank document or a **template** (report, letter, résumé).
- **Save** (Ctrl + S) keeps changes; **Save As** creates a copy with a new name, format or location. Save often, or save to OneDrive with AutoSave on.
- **Views** — Print Layout (as it will print), Read Mode, Web Layout, Outline and Draft.

2.2 Editing

- **Select** text by dragging, double-click (a word) or triple-click (a paragraph).
- **Cut, copy and paste** (Ctrl + X, C, V) move or duplicate text; **Undo/Redo** (Ctrl + Z, Y) reverse mistakes.
- **Find and Replace** (Ctrl + H) changes every occurrence at once — e.g. "fertiliser" to "fertilizer".
- **Spelling and Grammar** (F7) checks the text; set the proofing language correctly.
- **Track Changes and Comments** (Review tab) let a supervisor suggest edits that you accept or reject — essential for project reports and theses.

2.3 Formatting

Level	What you can set
Character	Font, size, bold, italic, underline, colour, superscript (m^2) and subscript (CO_2)
Paragraph	Alignment (left, centre, right, justify), line spacing, spacing before/after, indents, bullets and numbering
Styles	Named sets of formatting — Normal, Heading 1, Heading 2 — that keep a long document consistent
Page	Margins, orientation (portrait/landscape), paper size, columns, page and section breaks, headers, footers and page numbers

> Apply **Heading styles** to every chapter and section title. Word can then build an automatic **table of contents** (References tab) and a clickable navigation pane, and update them in one click when the report changes.

2.4 Tables, pictures and equations

- **Tables** (Insert → Table) present data in rows and columns; you can merge cells, add borders and repeat the header row on each page.
- **Pictures, shapes, SmartArt and charts** illustrate the text; add a **caption** (References → Insert Caption) so figures are numbered automatically.
- **Equations** (Insert → Equation) type formulas such as Q = CIA/360 properly.

2.5 References and citations

- **Footnotes and endnotes** add notes at the bottom of the page or the end of the document.
- **Citations and Bibliography** (References tab) store sources and format the reference list in a chosen style such as APA.
- **Cross-references** link to numbered figures and tables, and update when numbers change.

2.6 Mail merge

Mail merge combines a standard letter with a list of names and details to produce personalised copies — for example, invitation letters to 200 farmers for a field day, or certificates for a training programme.

1. Prepare the **data source** — an Excel sheet with columns such as Name, Village and Mobile.
2. Write the **main document** in Word (Mailings tab → Start Mail Merge → Letters).
3. Insert **merge fields** («Name», «Village») where the details should appear.
4. **Preview** the results, then **Finish & Merge** to print or create one document per farmer.

2.7 Indian-language typing

Word uses Unicode (Lecture 3), so documents can be written in Hindi and other Indian languages. Add the language keyboard in the operating system settings and use a Unicode font such as Nirmala UI or Mangal so the text displays correctly on other computers.

3. MS Excel — spreadsheets

3.1 Workbook, worksheet and cell

- A **workbook** (.xlsx file) contains one or more **worksheets** (tabs).
- Each worksheet is a grid of **columns** (A, B, C …) and **rows** (1, 2, 3 …). A modern worksheet has 1,048,576 rows and 16,384 columns (A to XFD) (Microsoft, n.d.-c).
- A **cell** is the meeting point of a column and a row, named by its **address** — B9 is column B, row 9. A **range** is a block of cells, such as B2:B8.
- The **formula bar** shows what a cell really contains — a value or a formula.

3.2 Data entry and formatting

- Enter **text** (labels), **numbers** and **dates** in separate cells — one value per cell, one record per row, one field per column.
- **Number formats** show currency (₹), percentages, decimals and dates without changing the stored value.
- **Conditional formatting** colours cells automatically — e.g. soil moisture below 35% shown in red.
- **Freeze panes** keep headings visible while scrolling; **data validation** restricts entries (e.g. only dates, or a drop-down list of crops).

3.3 Formulas

A **formula** always begins with = and can use numbers, cell addresses, operators and functions. Excel follows the usual order of operations: brackets, then ^ (power), then * and /, then + and −. For example, =2+3*4 gives 14, but =(2+3)*4 gives 20.

fx B9 =SUM(B2:B8)

	A	B	C
1	**Item (wheat, 1 ha)**	**₹**	**Formula shown**
2	Seed	5,000	
3	Fertilisers	7,500	
4	Plant protection	2,500	
5	Irrigation	4,000	
6	Machinery hire	9,000	
7	Hired labour	6,000	
8	Interest and misc.	2,000	
9	**Total cost**	**36,000**	**=SUM(B2:B8)**
10	Yield (q/ha)	45	
11	Price (₹/q)	2,400	
12	Gross return	1,08,000	=B10*B11
13	Net return	72,000	=B12−B9
14	**Benefit–cost ratio**	**3.0**	**=B12/B9**

Figure 2. A cost-of-cultivation worksheet. Column C shows the formulas used in column B. Prices are illustrative. Original diagram.

Worked example 1 — cost of cultivation and benefit–cost ratio (illustrative figures for 1 ha of wheat)
Total cost B9: =SUM(B2:B8) = 5,000 + 7,500 + 2,500 + 4,000 + 9,000 + 6,000 + 2,000 = ₹36,000
Gross return B12: =B10*B11 = 45 q × ₹2,400 = ₹1,08,000
Net return B13: =B12−B9 = ₹72,000
Benefit–cost ratio B14: =B12/B9 = 108,000 ÷ 36,000 = 3.0
Change any cost or the price and every result updates at once — the main strength of a spreadsheet.

3.4 Relative and absolute references

- A **relative** reference (A1) changes when the formula is copied: =B2*2 copied down becomes =B3*2, =B4*2 …
- An **absolute** reference (A1) stays fixed when copied — use it for constants such as a conversion factor or a price.
- A **mixed** reference ($A1 or A$1) fixes only the column or only the row. Press F4 to cycle through the types.

Worked example 2 — acres to hectares with an absolute reference
Cell E1 holds the factor 0.4047 (1 acre = 0.4047 ha). Field areas in acres are in B2:B4 (2, 3.5 and 5).
In C2 type =B2*E1 and copy it down to C4: Excel writes =B3*E1 and =B4*E1.
Results: 0.809, 1.416 and 2.024 ha. Without the $ signs, the copied formulas would point to E2 and E3, which are empty, and give 0.

3.5 Useful functions

Function	Purpose	Example
SUM, AVERAGE, MIN, MAX	Total, mean, smallest and largest value	=AVERAGE(B2:B11)
COUNT, COUNTIF	Count numbers; count cells meeting a condition	=COUNTIF(B2:B11,">=2.5")
SUMIF	Add values meeting a condition	=SUMIF(A2:A50,"Wheat",C2:C50)
IF	Choose a result by a test	=IF(C2<35,"Irrigate","OK")
ROUND	Round to given decimals	=ROUND(B2*E1,2)
VLOOKUP, XLOOKUP	Find a value in a table	=XLOOKUP("Maize",A2:A20,B2:B20)
TODAY	Current date	=TODAY()-B2 gives days after sowing
STDEV.S, CORREL	Standard deviation; correlation coefficient	=CORREL(B2:B20,C2:C20)
SLOPE, INTERCEPT, RSQ	Straight-line fit y = mx + c and its R^2	=SLOPE(C2:C5,B2:B5)

XLOOKUP is available in Excel 2021, Excel 2024 and Microsoft 365, but not in Excel 2016 or 2019, where VLOOKUP is used instead (Microsoft, n.d.-e).

> **Worked example 3 — rainfall summary**
> Daily rainfall (mm) for 10 days in B2:B11: 0, 3.2, 12.5, 0, 0.8, 25.0, 2.5, 0, 6.4, 1.2
> Total =SUM(B2:B11) = 51.6 mm · Mean =AVERAGE(B2:B11) = 5.16 mm · Maximum =MAX(B2:B11) = 25.0 mm
> Rainy days =COUNTIF(B2:B11,">=2.5") = 5 (3.2, 12.5, 25.0, 2.5 and 6.4 mm). The India Meteorological Department counts a day as rainy when it receives 2.5 mm or more (India Meteorological Department, n.d.).

3.6 Sorting, filtering and pivot tables

- **Sort** arranges rows by one or more columns (e.g. yield, highest first). **Filter** shows only the rows you want (e.g. one village).
- A **pivot table** summarises large data quickly — e.g. total area and average yield of each crop in each district — by dragging fields into rows, columns and values.

3.7 Charts and graphs

Purpose	Best chart	Agricultural example
Compare categories	Column or bar	Yield of five varieties
Show change over time	Line (or column for daily totals)	Monthly rainfall, mandi prices over a season
Show parts of a whole	Pie (few slices) or stacked column	Share of each cost item
Show a relationship	Scatter (XY), with a trendline	Sensor reading vs soil moisture; fertiliser dose vs yield
Show a distribution	Histogram	Spread of plot yields in a trial

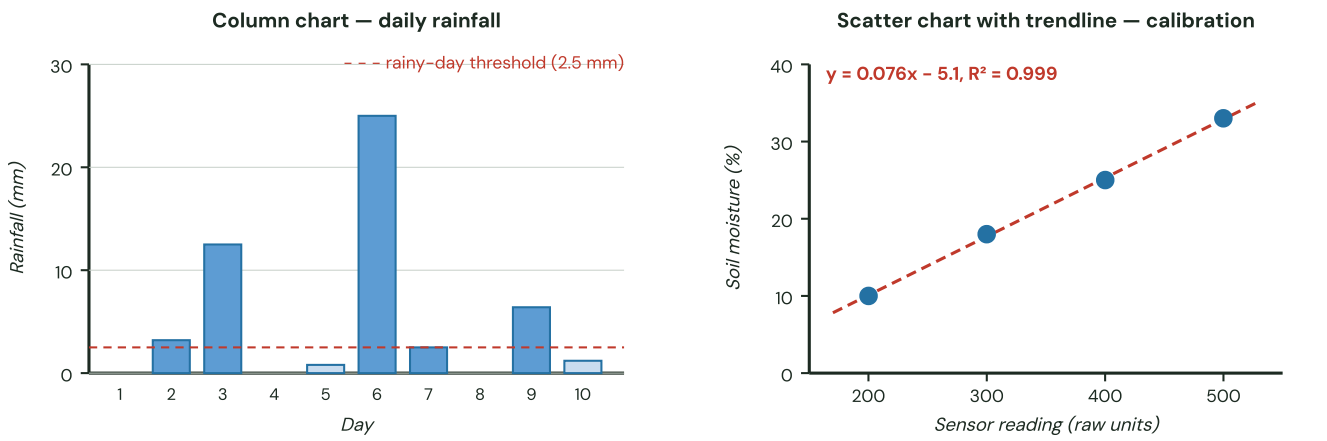


Figure 3. Left: daily rainfall from worked example 3, with rainy days (≥ 2.5 mm) darker. Right: sensor calibration from worked example 4. Original charts.

Worked example 4 — calibrating a soil-moisture sensor with a trendline
Raw sensor readings 200, 300, 400 and 500 were matched with laboratory (gravimetric) moisture of 10, 18, 25 and 33%.
Insert a **scatter chart**, add a **linear trendline** and tick "Display equation" and "Display R-squared".
Excel shows y = 0.076x - 5.1 and R² = 0.999 (the same values come from =SLOPE, =INTERCEPT and =RSQ).
The sensor can now be read as: moisture (%) = 0.076 × reading - 5.1. A reading of 350 gives 0.076 × 350 - 5.1 = 21.5%.

Good charts: give a clear title; label both axes with units; start column and bar axes at zero; avoid 3-D effects; use few colours; and state the data source.

4. MS PowerPoint — presentations

Feature names follow Microsoft's PowerPoint help pages (Microsoft, n.d.-b).

4.1 Building a presentation

- A presentation is a set of **slides**. Choose a **layout** for each slide (title, title and content, two content, picture with caption) and a **theme** (design and colours).
- Insert text, pictures, **tables**, **charts** (linked to Excel data), **SmartArt** diagrams, audio and video.
- The **slide master** sets fonts, colours and the logo for every slide at once.
- **Transitions** (between slides) and **animations** (within a slide) should be simple and few — they should help the message, not distract.
- **Views:** Normal (editing), Slide Sorter (reordering), Notes Page, Reading View and Slide Show. **Presenter View** shows your speaker notes and the next slide on your laptop while the audience sees only the slide.
- Save as .pptx and also export a **PDF** copy in case the venue's computer lacks the right fonts or software.

4.2 Designing a clear presentation

- One main idea per slide, with a headline that states the point ("Drip irrigation saved 40% water" rather than "Results").
- Few words: key phrases, not paragraphs. Large text — body text that can be read from the back of the room (around 24 pt or larger is a common guideline).

- Strong contrast (dark text on a light background, or the reverse); avoid red–green combinations.
- Prefer photos, charts and diagrams over text; label charts clearly.
- For farmers' meetings, use the local language, field photos and simple numbers; check the projector and lighting beforehand.
- End with a summary slide and credit all sources and photographs.

5. Working together and staying safe

- **Cloud storage and co-authoring** (OneDrive, SharePoint or Google Drive) let several people edit one file and keep a **version history**.
- **Protect** files with passwords where needed, and keep backups (Lecture 3).
- Use licensed or free software — pirated copies often carry malware (Lecture 4).

Shortcut	Action	Shortcut	Action
Ctrl + N	New file	Ctrl + B / I / U	Bold / italic / underline
Ctrl + O	Open	Ctrl + F / H	Find / replace
Ctrl + S	Save	Ctrl + A	Select all
Ctrl + P	Print	Ctrl + Enter	Page break (Word)
Ctrl + C / X / V	Copy / cut / paste	F4	Change reference type (Excel)
Ctrl + Z / Y	Undo / redo	F5	Start slide show (PowerPoint)

6. Key takeaways

- Word creates and formats documents; styles, captions and citations make long reports consistent and easy to update.
- Mail merge produces personalised letters and certificates from an Excel list.
- In Excel every formula starts with =; relative references change when copied, absolute references (A1) do not.
- SUM, AVERAGE, MAX, COUNTIF, IF and XLOOKUP handle most farm-record calculations.
- Choose charts by purpose; scatter charts with trendlines turn calibration data into equations.
- PowerPoint presentations work best with one idea per slide, large text, strong contrast and good visuals.
- .docx, .xlsx and .pptx are open standards (ISO/IEC 29500); LibreOffice and Google tools are free alternatives.

7. Examination questions

Short-answer questions (2 marks)

Q1. What is the difference between Save and Save As?

Save stores changes to the current file; Save As makes a copy with a new name, format or location.

Q2. What is mail merge? Give an agricultural use.

Combining a standard document with a data list to produce personalised copies — e.g. invitation letters to farmers for a field day.

Q3. Why should Heading styles be used in Word?

They keep formatting consistent and allow an automatic, updatable table of contents and navigation pane.

Q4. What is a cell address? Give an example.

The column letter and row number of a cell — e.g. B9.

Q5. Distinguish between relative and absolute cell references.

A relative reference (A1) changes when copied; an absolute reference (A1) stays fixed.

Q6. Write an Excel formula to count days with rainfall of 2.5 mm or more in B2:B32.

=COUNTIF(B2:B32,">=2.5")

Q7. What does =IF(C2<35,"Irrigate","OK") do?

It shows "Irrigate" when the value in C2 is below 35, otherwise "OK".

Q8. Which chart is best for showing a relationship between two variables?

A scatter (XY) chart, usually with a trendline.

Q9. What is the slide master in PowerPoint?

The master slide that sets the design, fonts, colours and logo for all slides.

Q10. Name the file extensions of Word, Excel and PowerPoint documents.

.docx, .xlsx and .pptx.

Long-answer questions (5–10 marks) — answer outlines

Q1. Describe the editing and formatting features of MS Word.

Selection, clipboard, undo, find and replace, spelling, track changes; character, paragraph, style and page formatting; tables, captions, references.

Q2. Explain mail merge with its steps.

Data source, main document, merge fields, preview, finish — with the farmers' field-day example.

Q3. Explain formulas and cell references in Excel with examples.

Formula rules and operator order; relative, absolute and mixed references; the acres-to-hectares example.

Q4. Prepare an Excel sheet for the cost of cultivation and benefit–cost ratio of a crop.

Layout of Figure 2 with SUM, product and ratio formulas; interpretation of B:C ratio.

Q5. Explain the types of charts in Excel and when to use each.

Column/bar, line, pie, scatter, histogram (table in Section 3.7); trendline and R^2 with the calibration example; good-chart rules.

Q6. Describe how to prepare an effective presentation in PowerPoint.

Layouts, themes, master, inserting objects, transitions, views, presenter view; design guidelines for seminars and farmers' meetings.

8. Glossary

Term	Meaning
Absolute reference	A cell reference with $ signs that does not change when copied
Cell	The box where a row and a column meet in a spreadsheet
Chart	A graphical display of data
Conditional formatting	Formatting applied automatically when cell values meet a condition
Formula	A calculation in a cell, beginning with =
Function	A built-in formula such as SUM or IF
Mail merge	Producing personalised copies of a document from a data list
Pivot table	An interactive summary table of large data
Presenter view	A PowerPoint view showing notes and the next slide to the speaker only
Range	A rectangular block of cells, such as B2:B8
Relative reference	A cell reference that adjusts when copied
Ribbon	The band of tabs and command groups at the top of Office programs
Slide master	The template slide controlling the design of all slides
Style	A named set of formatting applied to text
Template	A ready-made document design to start from
Track changes	A Word feature that records edits for review
Trendline	A best-fit line added to a chart
Workbook	An Excel file containing one or more worksheets

References

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All diagrams and charts are original. The cost, rainfall and calibration data are illustrative values prepared for teaching, not field results. Office features are described for current Microsoft 365 versions; menus differ slightly in older versions and in LibreOffice or Google tools.

Practice MCQs

Q1. A formula in Excel must begin with

(a) +
(b) =
(c) #
(d) @

Q2. In Excel, =2+3*4 returns

(a) 20
(b) 14
(c) 24
(d) 9

Q3. Which is an absolute cell reference?

(a) A1
(b) $A1
(c) A$1
(d) A1

Q4. Copying =B2*E1 from C2 to C3 gives

(a) =B2*E1
(b) =B3*E1
(c) =B3*E2
(d) =C3*E1

Q5. The formula to count cells in B2:B11 with values of 2.5 or more is

(a) =COUNT(B2:B11)
(b) =SUMIF(B2:B11,">=2.5")
(c) =COUNTIF(B2:B11,">=2.5")
(d) =IF(B2:B11>=2.5)

Q6. The best chart to show the relationship between fertiliser dose and yield is

(a) pie
(b) column
(c) scatter (XY)
(d) doughnut

Q7. A modern Excel worksheet has how many rows?

(a) 65,536
(b) 1,048,576
(c) 16,384
(d) 10,00,000

Q8. Personalised letters to many farmers are produced in Word using

(a) track changes
(b) mail merge
(c) styles
(d) macros

Q9. A Word feature that records edits for review is

(a) Find and Replace
(b) Track Changes
(c) Mail Merge
(d) Format Painter

Q10. The PowerPoint view that shows speaker notes only to the presenter is

(a) Slide Sorter
(b) Reading View
(c) Presenter View
(d) Outline View

Q11. The open standard behind .docx, .xlsx and .pptx files is

(a) ISO 9001
(b) ISO/IEC 29500
(c) ISO 32000
(d) ISO 14001

Q12. Total cost ₹36,000 and gross return ₹1,08,000 give a benefit–cost ratio of

(a) 0.33
(b) 2.0
(c) 3.0
(d) 72,000

Answers

Q1 — (b) =

Q2 — (b) 14 — *Multiplication is done before addition.*

Q3 — (d) A1

Q4 — (b) =B3*E1

Q5 — (c) =COUNTIF(B2:B11,">=2.5")

Q6 — (c) scatter (XY)
Q7 — (b) 1,048,576
Q8 — (b) mail merge
Q9 — (b) Track Changes
Q10 — (c) Presenter View
Q11 — (b) ISO/IEC 29500
Q12 — (c) 3.0